

Impact of VAT Adoption on Child Health outcomes in India

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ABSTRACT

This study examines whether the staggered adoption of Value Added Tax (VAT) across Indian states affected child mortality outcomes. Using child-level data from the National Family Health Survey (NFHS-4) and household consumption data from the National Sample Survey (NSS) Consumer Expenditure Surveys, this paper estimates the effect of VAT exposure on infant mortality and under-five mortality. The empirical strategy exploits variation in the timing of VAT adoption across Indian states using a difference-in-differences fixed effects framework. To examine the mechanism through which VAT may affect child health, the paper also estimates the effect of VAT adoption on monthly per capita consumption expenditure (MPCE). Across all baseline specifications, VAT exposure is associated with statistically insignificant effects on both child mortality outcomes and household consumption expenditure. These findings remain robust after incorporating household controls, state-specific time trends, and mother fixed effects. The results suggest that while VAT represented a significant fiscal reform, it did not produce measurable effects on the welfare outcomes examined in this study.

1. INTRODUCTION

Tax reforms in developing economies are often evaluated in terms of their effects on revenue generation, efficiency, and market formalization (Keen, 2008; Fullerton and Metcalf, 2002). However, fiscal reforms may also generate important welfare consequences for households, particularly in contexts where a large share of household income is spent on essential goods such as food, healthcare, and basic consumption items. Despite extensive research on taxation and public finance, relatively limited attention has been given to how indirect tax reforms affect household welfare outcomes.

Indirect taxes may influence household welfare through changes in consumer prices and purchasing power. Existing literature on tax incidence suggests that taxes imposed on producers are often partially or fully passed on to consumers through higher prices (Poterba, 1996; Fullerton and Metcalf, 2002). As a result, households, especially lower-income households, may experience reductions in real consumption expenditure following tax reforms. In developing countries, where many households face liquidity constraints and limited consumption-smoothing capacity, such reductions may have broader implications for nutrition, healthcare spending, and child welfare (Emran and Stiglitz, 2005).

Child health outcomes are particularly important indicators of household welfare. Previous research has shown that household economic conditions play a major role in determining child survival outcomes, especially in low and middle-income countries (Strauss and Thomas, 1998; Cutler et al., 2006). A decline in household consumption expenditure may reduce access to adequate nutrition, healthcare services, and other inputs essential for child development and survival. However, while substantial literature examines determinants of child mortality, the role of fiscal policy and indirect taxation remains relatively underexplored.

This paper examines whether the adoption of Value Added Tax (VAT) across Indian states affected household welfare outcomes, with a particular focus on child mortality. Specifically, the study investigates whether exposure to VAT influenced infant mortality and under-five mortality outcomes using data from the National Family Health Survey. To examine the po-

tential mechanism underlying these effects, the paper also analyzes the relationship between VAT adoption and household consumption expenditure using data from the NSS Consumer Expenditure Surveys.

The empirical analysis exploits variation in the timing of VAT adoption across Indian states, drawing on the state-level adoption data documented by Agrawal and Zimmermann (2024). Using a difference-in-differences fixed effects framework, the study compares outcomes across states before and after VAT implementation while controlling for state-specific and time-specific factors. The findings suggest that VAT adoption was associated with a small, negative, but statistically insignificant effect on household consumption expenditure and child health outcomes. These findings contribute to the literature on taxation and welfare by suggesting that the broader social consequences of VAT adoption in India were more limited than theoretical concerns regarding indirect taxation and household welfare would predict (Emran and Stiglitz, 2005; Fullerton and Metcalf, 2002).

The remainder of the paper is organized as follows. Section 2 discusses the background and implementation of VAT in India. Section 3 reviews the relevant literature. Section 4 describes the data sources. Section 5 outlines the empirical strategy. Section 6 presents the results, and Section 7 concludes.

2. BACKGROUND

India's indirect tax system prior to the introduction of Value Added Tax (VAT) was characterized by a complex and fragmented structure of state-level sales taxes. Under this regime, taxes were levied at multiple points along the production and distribution chain without any provision for firms to claim credits on taxes already paid on intermediate inputs. This cascading effect meant that taxes were effectively imposed on previously taxed value, compounding at each stage of production and artificially inflating final consumer prices (Keen, 2008). This resulted in a system that was both economically distortionary and administratively burdensome, which created incentives for tax evasion and reduced overall market efficiency (Emran and Stiglitz, 2005).

The pre-VAT tax architecture also suffered from significant inter-state variation. Each state maintained its own schedule of tax rates, exemptions, surcharges, and administrative procedures, leading to wide disparities in the effective tax burden across jurisdictions. Firms operating across multiple states faced major compliance costs due to this heterogeneity, and the lack of uniformity in tax structures created opportunities for tax arbitrage and distorted the geographic allocation of economic activity (Agrawal and Zimmermann, 2024). The complexity of the system also made it difficult for tax authorities to effectively monitor compliance, contributing to chronic problems of tax evasion and revenue leakage (Keen, 2008).

Recognizing these structural deficiencies, the Government of India, in coordination with the Empowered Committee of State Finance Ministers, undertook a comprehensive reform of the indirect tax system through the introduction of state-level VAT beginning in April 2005. The VAT system was designed to replace the cascading sales tax regime with a tax levied only on the value added at each stage of production, while allowing firms to claim input tax credits for taxes already paid on purchases of intermediate goods (Keen, 2008). This mechanism effectively eliminated the cascading effect, reducing the overall tax burden on production, lowering distortions in relative prices, and improving the transparency and administrative efficiency of the tax system. By creating a paper trail of transactions through the invoice-based credit mechanism, VAT also enhanced the ability of tax authorities to detect evasion and improve compliance

(Agrawal and Zimmermann, 2024).

Although the majority of Indian states adopted VAT in April 2005, the transition was not simultaneous across all states. Several states, including Uttar Pradesh, Rajasthan, and a few others, delayed adoption due to political and administrative considerations, with some states implementing the reform as late as 2008 (Agrawal and Zimmermann, 2024). This staggered rollout created a natural experiment in which different states were exposed to the VAT regime at different points in time, generating variation in VAT exposure that is exploited in the empirical analysis of this paper. The variation in adoption timing is plausibly exogenous to household welfare outcomes, as it was primarily driven by state-level political economy considerations and the pace of administrative preparation rather than by contemporaneous changes in household consumption or child health conditions.

The introduction of VAT represented one of the most consequential indirect tax reforms in India since independence, laying the institutional and administrative groundwork for the subsequent nationwide implementation of the Goods and Services Tax (GST) in 2017. While the primary motivation for the reform was fiscal to improve revenue efficiency, reduce tax-induced distortions, and broaden the tax base (Keen, 2008), the reform also had the potential to generate significant consequences for households through its effects on consumer prices. Standard economic theory predicts that indirect taxes are partially or fully shifted forward onto consumers through higher prices, depending on the elasticity of demand and the degree of market competition (Poterba, 1996; Fullerton and Metcalf, 2002). To the extent that VAT adoption altered the effective tax burden on final goods, it may have changed the relative prices faced by households, with potentially important implications for real purchasing power and consumption expenditure.

These price effects are of particular policy relevance in the Indian context, where a substantial share of household expenditure is devoted to food and other essential goods that are consumed disproportionately by lower-income households (Fullerton and Metcalf, 2002). If VAT adoption raised the prices of such goods, the reform could have generated regressive welfare consequences, reducing the real consumption expenditure of poorer households and

potentially affecting their ability to invest in child health and nutrition (Strauss and Thomas, 1998). Conversely, if VAT reduced cascading taxes and led to lower final prices for some goods, or if the reform had minimal pass-through to consumer prices due to market structure or informal sector dynamics (Emran and Stiglitz, 2005), its welfare consequences may have been limited. The empirical question of how VAT adoption affected household consumption expenditure and child health outcomes is therefore an important one, and this is what motivates the analysis in this paper.

3. LITERATURE REVIEW

This paper draws on and contributes to three distinct but related strands of the economics literature: the literature on the adoption and economic effects of Value Added Tax, the literature on tax incidence and household welfare, and the literature on the determinants of child health and mortality in developing countries. While each of these bodies of work is extensive, they have largely developed in isolation from one another, and limited research has examined how indirect tax reforms may affect household welfare through changes in consumption and health outcomes. This paper bridges these literatures by providing empirical evidence on the welfare consequences of VAT adoption in India, particularly its impact on child health outcomes.

The first strand of literature examines the adoption, design, and economic efficiency of Value Added Tax reform. VAT has been one of the most widely adopted fiscal innovations of the twentieth century, spreading from its origins in France in the 1950s to over 160 countries by the early 2000s. Keen (2008) provides a comprehensive analysis of VAT in developing countries, arguing that VAT represents a significant improvement over traditional sales tax systems due to its ability to reduce cascading taxation, improve revenue collection, and enhance administrative transparency through the invoice-based credit mechanism. However, Keen (2008) also notes that the efficiency gains from VAT may be partially offset in economies with large informal sectors, where the credit mechanism breaks down at the boundary between formal and informal firms. This observation is particularly relevant in the Indian context, where informal economic activity accounts for a substantial share of total output and employment (Emran and Stiglitz, 2005).

In the Indian context specifically, the staggered adoption of state-level VAT beginning in 2005 has attracted growing empirical attention. Agrawal and Zimmermann (2024) provide the most comprehensive recent analysis of the effects of VAT adoption in India, examining its consequences for firm-level outcomes including registration, output, and tax compliance. Their study finds that VAT adoption induced significant changes in firm behavior, particularly among smaller firms at the margin of the formal sector. Their replication dataset, which documents

the precise timing of VAT adoption across Indian states, forms the basis for the VAT exposure variable used in the present study. By extending the analysis from firm-level to household-level outcomes, this paper complements Agrawal and Zimmermann (2024) and contributes to a more complete understanding of the broader economic consequences of the VAT reform in India.

The second strand of relevant literature concerns tax incidence and the distributional consequences of indirect taxation. A central question in public finance is the extent to which taxes imposed on producers are ultimately borne by consumers through higher prices. Poterba (1996) provides influential evidence from the United States showing that changes in state and local sales taxes are largely passed through to retail prices, with estimated pass-through rates close to or exceeding one. This finding suggests that indirect taxes are borne primarily by consumers rather than producers, at least in competitive retail markets. Fullerton and Metcalf (2002) offer a comprehensive theoretical treatment of tax incidence, showing that the distribution of the tax burden across households depends critically on the relative shares of factor income and consumption goods, and that indirect taxes on necessities tend to be regressive because poorer households devote a larger share of their budgets to such goods.

These insights are particularly salient in the context of developing countries like India. Emran and Stiglitz (2005) argue that the welfare effects of indirect tax reform in economies with large informal sectors are theoretically ambiguous and may differ substantially from predictions based on models of fully formal economies. In such settings, the introduction of VAT may shift the tax burden toward formal sector goods, potentially increasing prices for consumers who purchase formal goods while leaving informal sector prices relatively unchanged (Emran and Stiglitz, 2005). The distributional consequences of such price changes depend on the income level and consumption patterns of affected households, making the empirical analysis of VAT's welfare effects particularly important. In the Indian context, where rural and lower-income households are heavily reliant on food expenditure and may have differential access to formal versus informal markets (Fullerton and Metcalf, 2002), these considerations are especially relevant.

Beyond tax incidence, a broader literature in development economics examines how

household income and consumption shocks affect welfare outcomes. Households in developing countries often face significant credit market imperfections that limit their ability to smooth consumption in response to adverse income or price shocks (Strauss and Thomas, 1998). As a result, reductions in real household purchasing power may translate directly into lower expenditure on food, healthcare, education, and other goods that are important for child health. The National Sample Survey consumption data used in this study have been extensively employed in the Indian development economics literature to examine how household welfare responds to economic shocks, policy reforms, and structural changes. This literature provides the conceptual and empirical foundation for using Monthly Per Capita Expenditure (MPCE) as a mechanism through which VAT adoption may affect child health outcomes (Strauss and Thomas, 1998).

The third strand of literature examines the determinants of child health and mortality in developing countries. A large body of evidence consistently identifies household socioeconomic status, maternal education, access to healthcare, nutritional intake, and sanitation conditions as the primary determinants of child survival outcomes. Strauss and Thomas (1998) provide an authoritative review of the relationship between health, nutrition, and economic development, documenting robust evidence that household income and consumption are positively associated with child health outcomes across a wide range of developing country contexts. Their analysis highlights that the relationship between income and health is not merely correlational but reflects genuine causal pathways through which economic resources shape access to nutrition, healthcare, and sanitation (Strauss and Thomas, 1998).

Cutler et al. (2006) extend this analysis to examine the determinants of mortality more broadly, finding that income growth, technological progress in medicine, and improvements in public health infrastructure have been the primary drivers of mortality decline over the long run. Importantly, their analysis suggests that income effects on mortality operate both directly, through households' ability to purchase healthcare and nutrition, and indirectly, through their effects on public health investment and sanitation infrastructure (Cutler et al., 2006). This broader perspective reinforces the theoretical plausibility of the mechanism examined in this paper: if VAT adoption reduces household real consumption, the effect may transfer to child

health outcomes through multiple channels, including reduced expenditure on food, reduced utilization of healthcare services, and reduced investment in child-specific goods (Strauss and Thomas, 1998; Cutler et al., 2006).

Despite the richness of these literatures, the intersection between fiscal policy and child health outcomes in developing countries remains largely unexplored. Studies of VAT and indirect taxation have focused primarily on efficiency and revenue outcomes (Keen, 2008; Agrawal and Zimmermann, 2024), while the child health literature has focused on determinants such as nutrition and healthcare access without adequately examining the role of fiscal policy as a distal determinant (Strauss and Thomas, 1998; Cutler et al., 2006). This paper addresses this gap by providing direct empirical evidence on whether VAT adoption in India affected household consumption expenditure and child mortality outcomes, thereby connecting the fiscal reform literature with the household welfare and child health literatures in a unified empirical framework.

4. DATA

This study combines child health data, household-level consumption data, and state-level policy information to examine the impact of VAT adoption on household welfare outcomes in India. The empirical analysis draws on three primary sources: the National Family Health Survey, the NSS Consumer Expenditure Surveys, and state-level VAT adoption data obtained from the replication files of Agrawal and Zimmermann (2024).

4.1 National Family Health Survey (NFHS-4)

To examine child health outcomes, this study uses data from NFHS-4 (2015–16), a nationally representative household survey that collects extensive information on fertility, mortality, nutrition, and healthcare outcomes across India. The survey includes complete birth histories for women aged 15–49, providing detailed information on the timing of births and child deaths. This information is used to construct individual-level measures of child mortality. The unit of observation in the health analysis is the child.

This study examines two key child health outcomes: infant mortality, defined as a binary variable equal to 1 if a child died between 0 and 11 months of age, and 0 otherwise; and under-five mortality, defined as a binary variable equal to 1 if a child died before reaching five years of age (0–59 months), and 0 otherwise. To ensure complete exposure to mortality risk, children born within one year of the survey date are excluded from the infant mortality analysis, and children born within 60 months of the survey date are excluded from the under-five mortality analysis. After applying these restrictions, the infant mortality sample consists of 892,856 children and the under-five mortality sample consists of 686,389 children.

The NFHS dataset also provides several demographic and socioeconomic characteristics used as control variables, including child gender, birth order, mother’s education, household wealth indicators, rural or urban residence, social group indicators, religion indicators, and state identifiers. Table 1 presents summary statistics for the key variables in the child health sample.

Table 1: Summary Statistics: Child Health Sample (NFHS-4)

Variable	Mean	Std. Dev.	N
<i>Panel A: Infant Mortality Sample</i>			
Infant Mortality (0–11 months)	0.045	(0.206)	892,856
Female Child	0.478	(0.500)	892,856
Birth Order	2.329	(1.395)	892,856
Wealth Quintile	1.855	(0.885)	892,856
Urban Household	0.291	(0.454)	892,856
Muslim Household	0.159	(0.366)	892,856
VAT Exposure	0.475	(0.499)	892,856
<i>Panel B: Under-5 Mortality Sample</i>			
Under-5 Mortality (0–59 months)	0.057	(0.233)	686,389
Female Child	0.478	(0.500)	686,389
Birth Order	2.379	(1.417)	686,389
Wealth Quintile	1.855	(0.885)	686,389
Urban Household	0.293	(0.455)	686,389
Muslim Household	0.158	(0.364)	686,389
VAT Exposure	0.315	(0.465)	686,389

Notes: The table reports the mean and standard deviation of variables used in the child mortality regressions. Data are from NFHS-4 (2015–16). Panel A excludes twin births and children born within 12 months of the survey date, yielding 892,856 observations. Panel B applies the same twin exclusion and excludes children born within 60 months of the survey date, yielding 686,389 observations. Female Child is a binary variable equal to 1 if the child is female. Birth Order is the child’s birth order within the household. Wealth Quintile ranges from 1 (poorest) to 5 (wealthiest). Urban Household is a binary variable equal to 1 if the household resides in an urban area. Muslim Household is a binary variable equal to 1 if the household is Muslim. VAT Exposure is a binary variable equal to 1 if the child’s state had adopted VAT in the birth year. Individual sampling weights applied throughout.

4.2 VAT Adoption Data

The key explanatory variable in this study is exposure to the implementation of VAT at the state level. Data on state-wise VAT adoption dates is obtained from the replication files of Agrawal and Zimmermann (2024), which documents the staggered rollout of VAT across Indian states. Using the state-level adoption years, a binary VAT exposure variable is constructed: a state is coded as 0 in all years prior to adopting VAT and as 1 in the year of adoption and all subsequent years. Formally:

$$\text{VATExposure}_{st} = \begin{cases} 1 & \text{if } t \geq \text{VAT adoption year in state } s \\ 0 & \text{otherwise} \end{cases}$$

The VAT exposure variable is merged with both the NSS and NFHS datasets using state identifiers and survey year information.

4.3 National Sample Survey (NSS) Consumer Expenditure Data

To examine the effect of VAT adoption on household consumption, this study uses data from the NSS Consumer Expenditure Surveys from three rounds: the 55th Round (1999–2000), the 61st Round (2004–05), and the 68th Round (2011–12). The 55th round provides a pre-reform baseline, the 61st round captures the period immediately surrounding VAT adoption, and the 68th round reflects household consumption outcomes following implementation. The pooled NSS sample consists of 193,120 households.

The primary consumption outcome is Monthly Per Capita Consumption Expenditure (MPCE), calculated as total household monthly consumption expenditure divided by household size. Nominal expenditure values are deflated using the Consumer Price Index for Agricultural Labourers (CPI-AL) to convert them into real terms. The empirical analysis primarily uses the logarithm of real MPCE to account for skewness in the expenditure distribution. Survey sampling weights are applied to ensure nationally representative estimates. The NSS dataset also provides household-level demographic controls including rural or urban sector, social group or caste category, religion, household size, and state identifiers. Table 2 presents summary statistics for the key variables in the household consumption sample.

Table 2: Summary Statistics: Household Consumption Sample (NSS Pooled)

Variable	Mean	Std. Dev.	N
Log Real MPCE	1.985	(0.676)	193,120
VAT Exposure	0.594	(0.491)	193,120
Rural Household	0.731	(0.443)	193,120
SC/ST Household	0.275	(0.447)	193,120
Muslim Household	0.115	(0.320)	193,120

Notes: The table reports the mean and standard deviation of variables used in the primary consumption regressions. Data are from the pooled NSS Consumer Expenditure Surveys (55th, 61st, and 68th rounds), comprising 193,120 households. Log Real MPCE is the natural logarithm of monthly per capita consumption expenditure deflated using the Consumer Price Index for Agricultural Labourers (CPI-AL). Rural is a binary variable equal to 1 if the household resides in a rural area. SC/ST is a binary variable equal to 1 if the household belongs to a Scheduled Caste or Scheduled Tribe. Muslim is a binary variable equal to 1 if the household follows Islam. VAT Exposure is a binary variable equal to 1 if the household's state had adopted VAT in the survey year. Survey sampling weights applied throughout.

5. ESTIMATION STRATEGY

This study exploits the staggered adoption of Value Added Tax (VAT) across Indian states as a source of quasi-experimental variation to identify the causal effect of tax reform on household welfare outcomes (Agrawal and Zimmermann, 2024). Since states adopted VAT at different points in time rather than simultaneously, the reform generates cross-state and over-time variation in VAT exposure that is well-suited to a difference-in-differences (DiD) framework (Angrist and Pischke, 2009). The core idea is to compare changes in outcomes in states that adopted VAT against changes in outcomes in states that had not yet adopted VAT over the same period, using the latter as the counterfactual for what would have happened in adopting states in the absence of the reform.

The core identifying assumption underlying this strategy is the parallel trends assumption: in the absence of VAT adoption, early- and late-adopting states would have followed similar trends in child health and household consumption outcomes (Angrist and Pischke, 2009). This assumption is plausible in the Indian context because the timing of VAT adoption across states was primarily driven by a centrally coordinated reform process, and was largely determined by administrative and political factors rather than by pre-existing differences in household welfare trajectories across states (Agrawal and Zimmermann, 2024).

To estimate the effect of VAT adoption on child health, the following DiD specification is used:

$$Y_{ist} = \beta \text{VATExposure}_{st} + \gamma_s + \delta_t + X'_{ist} \theta + \varepsilon_{ist} \quad (1)$$

where Y_{ist} is a binary indicator equal to 1 if child i , born in state s in year t , died within the relevant mortality risk window (0–11 months for infant mortality; 0–59 months for under-five mortality), and 0 otherwise. VATExposure_{st} is the key explanatory variable, equal to 1 if state s had adopted VAT by year t and 0 otherwise, and the coefficient β is the DiD estimator of interest, capturing the average change in child mortality outcomes associated with VAT adoption relative to the counterfactual trend in non-adopting states (Angrist and Pischke, 2009). γ_s denotes state fixed effects, which absorb all time-invariant differences across states such as base-

line healthcare infrastructure, geographic characteristics, and long-run economic conditions. δ_t denotes birth year fixed effects, which absorb common year-specific shocks affecting child mortality uniformly across all states, such as national health policy changes or macroeconomic fluctuations. X_{ist} is a vector of child and household-level controls including child gender, birth order, mother's years of education, household wealth index, rural or urban residence, religion, and social group (Strauss and Thomas, 1998; Cutler et al., 2006). ϵ_{ist} is the error term.

The theoretical channel through which VAT adoption may affect child health operates through changes in household consumption expenditure (Strauss and Thomas, 1998). If VAT adoption raises consumer prices - particularly for essential goods such as food and healthcare - it may reduce real household purchasing power, which in turn constrains spending on child nutrition and preventive healthcare, thereby increasing child mortality risk (Cutler et al., 2006; Emran and Stiglitz, 2005). To examine this mechanism, an analogous DiD specification is estimated using household-level consumption data from the pooled NSS Consumer Expenditure Surveys:

$$\ln(\text{MPCE}_{ist}) = \beta \text{VATExposure}_{st} + \gamma_s + \delta_t + X'_{ist} \theta + \epsilon_{ist} \quad (2)$$

where $\ln(\text{MPCE}_{ist})$ is the logarithm of real monthly per capita consumption expenditure for household i in state s surveyed in period t , and γ_s and δ_t now denote state and survey round fixed effects respectively. The coefficient β captures the average percentage change in real MPCE associated with VAT adoption (Poterba, 1996).

To assess the robustness of the baseline estimates, several additional specifications are estimated. State-specific linear time trends are added to control for the possibility that adopting and non-adopting states were on divergent pre-reform trajectories. Mother fixed effects are incorporated into the child mortality regressions to exploit within-mother variation across births, comparing mortality outcomes for siblings born before and after VAT adoption within the same household and thereby controlling for all time-invariant unobserved household-level characteristics (Strauss and Thomas, 1998).

All specifications are estimated using Ordinary Least Squares (OLS) via the `reghdfe` command in Stata, which efficiently absorbs multiple high-dimensional fixed effects. A lin-

ear probability model is used for the binary mortality outcomes for ease of interpretation and consistency with the fixed effects structure (Angrist and Pischke, 2009). Standard errors are clustered at the state level throughout to account for within-state serial correlation in outcomes over time, which is the relevant level of treatment assignment in this setting.

6. RESULTS

6.1 Effect of VAT Exposure on Child Mortality

Table 3 (see Appendix) presents the baseline results for the effect of VAT exposure on under-five mortality. Across all four model specifications, ranging from a baseline model with only state and birth year fixed effects to a fully saturated model with controls and state-specific time trends, the coefficient on VAT exposure remains small, negative, and statistically insignificant. The estimates are stable as controls and state trends are added, suggesting the findings are not sensitive to alternative specifications.

Table 4 (see Appendix) presents the corresponding results for infant mortality. Consistent with the under-five mortality results, the coefficient on VAT exposure is small and statistically insignificant across all specifications. Although the sign and magnitude of the coefficient vary slightly across models, the results consistently fail to reject the null hypothesis that VAT adoption had a measurable effect on infant mortality. Taken together, these findings suggest that the introduction of VAT did not significantly alter child mortality outcomes in India during the study period.

6.2 Effect of VAT Exposure on Household Consumption

Table 5 (see Appendix) presents the results for the effect of VAT exposure on log real MPCE using the pooled NSS sample of 193,120 households. Across all specifications, VAT exposure has a negative but statistically insignificant effect on household consumption expenditure. The magnitude of the coefficient becomes slightly larger after adding household controls and state trends, but the overall conclusion remains unchanged. The absence of a significant consumption effect provides a plausible explanation for the null results in the child mortality regressions: if VAT adoption did not substantially reduce household purchasing power, its impact on child health outcomes would likely also be limited.

6.3 Robustness Checks

Table 6 (see Appendix) presents robustness checks incorporating mother fixed effects and state-specific time trends for both child mortality outcomes. The inclusion of mother fixed effects allows comparisons within households by controlling for unobserved household-level characteristics that remain constant across births. The coefficients on VAT exposure remain negative and statistically insignificant after incorporating these additional controls, confirming that the main findings are robust to within-family comparisons and stricter identification requirements.

7. CONCLUSION

This paper examined whether the staggered adoption of Value Added Tax (VAT) across Indian states affected household welfare, with a focus on child mortality outcomes and household consumption expenditure as the proposed transmission mechanism. Exploiting quasi-experimental variation in the timing of VAT adoption across states within a difference-in-differences framework, the analysis uses child-level data from NFHS-4 and household-level data from the pooled NSS Consumer Expenditure Surveys to estimate the effect of VAT exposure on infant mortality, under-five mortality, and monthly per capita consumption expenditure.

The main finding of the paper is that VAT adoption had no statistically significant effect on either infant mortality or under-five mortality across any of the specifications estimated. The coefficients on VAT exposure are consistently small in magnitude and precisely estimated, the results strongly support the null hypothesis - that VAT has no effect on child health. Consistent with these null results for child health, the analysis also finds no statistically significant effect of VAT adoption on household consumption expenditure. The coefficient on VAT exposure in the consumption regressions is negative across all specifications, suggesting a possible reduction in real MPCE following reform, but the estimates remain statistically insignificant even after the inclusion of household controls and state-specific time trends. These findings hold up across a range of robustness checks, including specifications with state-specific linear trends and mother fixed effects as well.

The absence of a significant consumption effect provides an internally coherent explanation for the null results in the child mortality regressions. The theoretical mechanism linking VAT adoption to child health operates through changes in household purchasing power: if VAT raises consumer prices, households experience a reduction in real consumption, which may in turn reduce expenditure on child nutrition, healthcare, and other welfare goods. The finding that VAT adoption did not significantly affect household consumption therefore implies that this transmission channel was not activated to a sufficient degree to generate measurable effects on child health outcomes. This may reflect several underlying factors. VAT may not have substantially altered the effective consumer prices of essential goods in India, either because

pass-through to consumer prices was limited due to market structure or informal sector dynamics, or because the reform reduced cascading taxes sufficiently to offset any upward price pressures. Alternatively, households may have partially adjusted their consumption behavior in response to price changes in ways that insulated child health outcomes from the effects of the reform.

These findings carry several important implications for the evaluation of indirect tax reforms in developing countries. First, they suggest that the welfare consequences of VAT adoption in India, at least with respect to child mortality and household consumption, were more limited than theoretical concerns about regressive indirect taxation might suggest. This is an important qualification to the view that indirect tax reforms necessarily impose welfare costs on poorer households through higher consumer prices. Second, the findings highlight the importance of examining welfare outcomes empirically rather than relying solely on theoretical predictions, as the actual incidence and welfare effects of tax reforms depend critically on context-specific factors such as market structure, the extent of informality, and the degree of tax pass-through to consumer prices.

At the same time, several important caveats qualify the interpretation of these findings. The difference-in-differences strategy relies on the parallel trends assumption, which, while plausible, cannot be directly tested due to data limitations on pre-reform trends in child health outcomes. If adopting and non-adopting states were on divergent welfare trajectories prior to VAT implementation for reasons unrelated to the reform, the estimates may be biased. Additionally, the analysis examines average effects across the full sample and may not capture important heterogeneity in the welfare consequences of VAT adoption across income groups, geographic regions, or rural and urban populations. It is possible that VAT adoption had significant adverse effects on specific vulnerable subgroups - such as the rural poor or households in states with weaker healthcare infrastructure - that are obscured by the average null effect estimated here. Furthermore, the long-run welfare consequences of VAT adoption may differ from the medium-run effects captured in this analysis, particularly if VAT generates gradual improvements in public service delivery through enhanced government revenue over time.

Future research could build on the findings of this paper in several directions. Examining heterogeneous treatment effects across income quintiles and geographic regions would provide a more deeper understanding of the distributional consequences of VAT adoption in India. Investigating the effect of VAT on the prices of specific consumption categories particularly food, healthcare, and education, would shed light on the precise mechanisms through which indirect tax reform affects household welfare. Event study methods could be used to examine pre-reform trends in outcomes and provide more direct evidence on the validity of the parallel trends assumption.

Overall, this paper contributes to a growing literature that evaluates fiscal reforms through the lens of household welfare and social outcomes. The results underscore that the welfare consequences of indirect tax reforms cannot be assumed a priori from theoretical considerations alone, and that careful empirical analysis using quasi-experimental methods is essential for understanding how fiscal policy affects the living standards of households in developing economies. While VAT represented one of the most significant fiscal reforms in post-independence India, the evidence from this study suggests that its effects on child health and household consumption were limited on the variables considered in this study.

REFERENCES

- Agrawal, D., and Zimmermann, L. (2024). The Effects of Adopting a Value Added Tax on Firms. *Working Paper*.
- Cutler, D., Deaton, A., and Lleras-Muney, A. (2006). The Determinants of Mortality. *Journal of Economic Perspectives*, 20(3), 97–120.
- Emran, M. S., and Stiglitz, J. E. (2005). On Selective Indirect Tax Reform in Developing Countries. *Journal of Public Economics*, 89(4), 599–623.
- Fullerton, D., and Metcalf, G. E. (2002). Tax Incidence. In A. J. Auerbach and M. Feldstein (Eds.), *Handbook of Public Economics*, Vol. 4 (pp. 1787–1872). Elsevier.
- Keen, M. (2008). VAT, Tariffs, and Withholding: Border Taxes and Informality in Developing Countries. *Journal of Public Economics*, 92(10–11), 1892–1906.
- Poterba, J. M. (1996). Retail Price Reactions to Changes in State and Local Sales Taxes. *National Tax Journal*, 49(2), 165–176.
- Strauss, J., and Thomas, D. (1998). Health, Nutrition, and Economic Development. *Journal of Economic Literature*, 36(2), 766–817.
- Angrist, J. D., and Pischke, J. S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.

APPENDIX

This appendix presents the full regression results referenced in Section 6 of the paper. Tables 3 and 4 report the baseline OLS estimates of the effect of VAT exposure on under-five mortality and infant mortality respectively. Table 5 reports the effect of VAT exposure on household consumption expenditure as the proposed mechanism. Table 6 presents robustness checks incorporating mother fixed effects.

Table 3: Effect of VAT Exposure on Under-Five Mortality

	(1)	(2)	(3)	(4)
VAT Exposure	−0.0004 (0.0014) [0.808]	−0.0021 (0.0013) [0.115]	0.0000 (0.0015) [0.995]	−0.0017 (0.0013) [0.166]
Observations	686,389	686,389	686,389	686,389
State FE	Yes	Yes	Yes	Yes
Birth Year FE	Yes	Yes	Yes	Yes
State Trends	No	Yes	No	Yes
Controls	No	No	Yes	Yes

Notes: Robust standard errors clustered at the state level in parentheses. p -values reported in square brackets. Controls include child, household, and demographic characteristics.

Table 4: Effect of VAT Exposure on Infant Mortality

	(1)	(2)	(3)	(4)
VAT Exposure	0.0001 (0.0015) [0.996]	0.0000 (0.0015) [0.996]	0.0004 (0.0014) [0.764]	0.0003 (0.0015) [0.817]
Observations	892,856	892,856	892,856	892,856
State FE	Yes	Yes	Yes	Yes
Birth Year FE	Yes	Yes	Yes	Yes
State Trends	No	Yes	No	Yes
Controls	No	No	Yes	Yes

Notes: Robust standard errors clustered at the state level in parentheses. p -values reported in square brackets. Controls include child, household, and demographic characteristics.

Table 5: Effect of VAT Exposure on Household Consumption (Log Real MPCE)

	(1)	(2)	(3)	(4)
VAT Exposure	−0.0164 (0.1476) [0.912]	−0.0709 (0.0979) [0.478]	−0.0946 (0.1571) [0.554]	−0.1048 (0.0750) [0.178]
Observations	193,120	193,120	186,312	186,312
State FE	Yes	Yes	Yes	Yes
Round FE	Yes	Yes	Yes	Yes
State Trends	No	Yes	No	Yes
Controls	No	No	Yes	Yes

Notes: Robust standard errors clustered at the state level in parentheses. p -values reported in square brackets. Controls include sector, social group, and religion.

Table 6: Robustness Check: Mother Fixed Effects

	Under-5 Mortality		Infant Mortality	
	(1)	(2)	(1)	(2)
VAT Exposure	−0.0018 (0.0012) [0.147]	−0.0020 (0.0013) [0.151]	−0.0006 (0.0012) [0.618]	−0.0005 (0.0014) [0.732]
Observations	581,032	581,032	774,446	774,446
Mother FE	Yes	Yes	Yes	Yes
Birth Year FE	Yes	Yes	Yes	Yes
State Trends	No	Yes	No	Yes
Controls	Yes	Yes	Yes	Yes

Notes: Robust standard errors clustered at the state level in parentheses. p -values reported in square brackets. Column (2) additionally includes state-specific linear trends.